Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2004-II-Q02] | Part A (Q1-Q5)

Answer: 441

1. The matching cases are RR/RR, BB/BB, and mixed/mixed. Each single-color case has probability $[\binom{10}{2}/\binom{20}{2}][\binom{8}{2}/\binom{18}{2}] = 14/323$.

2. The mixed case has probability $[100/\binom{20}{2}][81/\binom{18}{2}] = 90/323$.

3. Adding gives $118/323$, already reduced, so the answer is 441.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2006-II-Q05] | Part A (Q1-Q5)

Answer: 029

1. Write exceptional probabilities $1/6 + t$ and $1/6 - t$ with $t > 0$. The sum-seven probability is $4/36 + 2[(1/6)^2 - t^2] = 1/6 - 2t^2$.

2. Equating gives $2t^2 = 1/288$, so $t = 1/24$ and $P(F) = 5/24$. The answer is 29.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-I-Q02] | Part A (Q1-Q5)

Answer: 144

1. The two same-color events are disjoint, giving probability $[4 \cdot 16 + 6N]/[10(16 + N)] = 29/50$.

2. Clearing denominators gives $320 + 30N = 464 + 29N$, hence $N = 144$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-II-Q02] | Part A (Q1-Q5)

Answer: 076

1. Let the triple-intersection probability be t. Then $t/(0.14+t)=1/3$, giving $t=0.07$.

2. The union has probability $3(0.1)+3(0.14)+0.07=0.79$, while A has probability $0.1+2(0.14)+0.07=0.45$.

3. The desired probability is $0.21/0.55=21/55$, giving 76.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2015-I-Q05] | Part A (Q1-Q5)

Answer: 341

1. Deal the ten distinct socks into five labeled daily pairs. Permuting day labels preserves uniformity, so the pattern mismatch, mismatch, match has the same probability as match, mismatch, mismatch.

2. The latter pattern has probability $(1/9)(6/7)(13/15)$. After the first match, four complete pairs remain among eight socks; after a mismatch, only two complete pairs remain among six socks.

3. The product is $26/315$, already reduced. The requested sum is 341.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2015-II-Q02] | Part A (Q1-Q5)

Answer: 025

1. Using a population scale of 100, the Latin groups have sizes $40, 24, 10, 2$, for a total of 76. These are proportional counts, so no assumption about actual school size is needed.

2. The desired conditional probability is $24/76 = 6/19$. Hence $m + n = 25$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2016-I-Q02] | Part A (Q1-Q5)

Answer: 071

1. Normalization gives $\Pr(k) = k/21$. By independence, outcome $(k, 7-k)$ has probability $k(7-k)/21^2$.

2. Summing the six possibilities gives $(6+10+12+12+10+6)/441 = 56/441 = 8/63$. Therefore $m+n = 71$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2016-II-Q02] | Part A (Q1-Q5)

Answer: 107

1. Let $p = \Pr(\text{Sunday rain} \mid \text{Saturday dry})$. Total probability gives $(2/5)(2p) + (3/5)p = 3/10$, hence $p = 3/14$.

2. The probability both days are dry is $(3/5)(11/14) = 33/70$. Its complement is $37/70$, giving $a + b = 107$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2017-II-Q02] | Part A (Q1-Q5)

Answer: 781

1. Team4 reaches the final with probability $4/5$. Its opponent is team2 with probability $2/5$ and team3 with probability $3/5$.

2. Conditioning on that opponent, the title probability is $(4/5)[(2/5)(2/3)+(3/5)(4/7)]=256/525$. The numerator is a power of2 and the denominator odd, so the reduced sum is 781.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2021-I-Q01] | Part A (Q1-Q5)

Answer: 097

1. Exactly one of races 2 through 6 must be a loss. If the loss is last, the path has four repeats and one switch, with probability $16/243$.

2. At any of the other four positions, there are two switches and three repeats, each path having probability $8/243$. Thus the total is $(16 + 4 \cdot 8)/243 = 16/81$, giving 97.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2022-II-Q02] | Part A (Q1-Q5)

Answer: 125

1. With probability $1/3$, C faces A immediately. Conditional on that pairing, the title chance is $(1/3)(3/4) = 1/4$.

2. For either other pairing, C wins the semifinal with probability $3/4$. In the final the opponent is A with probability $3/4$, so the conditional title chance is $(3/4)[(3/4)(1/3) + (1/4)(3/4)] = 21/64$.

3. Average over pairings: $(1/3)(1/4) + (2/3)(21/64) = 29/96$. Thus $p + q = 125$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2024-I-Q04] | Part A (Q1-Q5)

Answer: 116

1. Within the prize event, draws with exactly $2, 3, 4$ matches number $\binom{4}{2}\binom{6}{2} = 90$, $\binom{4}{3}\binom{6}{1} = 24$, and 1.

2. Exactly one of these 115 equally likely draws is the top prize, so the conditional probability is $1/115$ and $m + n = 116$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-II-Q06] | Part B (Q6-Q10)

Answer: 167

1. Bayes gives posterior odds biased:fair equal to $(2/3)^2 : (1/6)^2 = 16 : 1$. Thus their posterior probabilities are $16/17, 1/17$.

2. The predictive probability is $(16/17)(2/3) + (1/17)(1/6) = 65/102$, already reduced.

3. The requested sum is 167.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2017-II-Q09] | Part B (Q6-Q10)

Answer: 013

1. Complete coverage by eight cards means exactly one repeated number and exactly one repeated color. A discard is possible exactly when their intersection card is present, in which case that card is uniquely discardable.

2. A favorable hand comes uniquely from a seven-card number-color matching (7! choices) plus one of the42 unused cards, yielding211680 hands.

3. For an unfavorable hand, choose the repeated number7 ways and its two colors $\binom{7}{2}$ ways. The repeated color must be among the other5; choose its two numbers from the remaining6 in $\binom{6}{2}$ ways, then match the four leftover numbers and colors in $4!$ ways. Count $7\binom{7}{2} \cdot 5\binom{6}{2} \cdot 4! = 264600$.

4. The conditional probability is $211680/(211680 + 264600) = 4/9$, giving $p + q = 13$.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2026-I-Q09] | Part B (Q6-Q10)

Answer: 029

1. Let f_j denote the face on roll j. Visibility requires $f_3, f_4, f_5, f_6 \neq f_2$ and $f_5, f_6 \neq f_4$. The conditioned sample space therefore has $6 \cdot 6 \cdot 5 \cdot 5 \cdot 4 \cdot 4 = 14400$ equiprobable sequences.

2. One blank face means five distinct rolled faces, so exactly one pair of roll indices coincides. Of the fifteen index pairs, four involving index two and later indices, and two involving four and later indices, are forbidden. Thus nine repeated-pair choices remain.

3. Assign five distinct faces in $(6)_5 = 720$ ways, giving 6480 favorable sequences. Probability $6480/14400 = 9/20$, and the requested sum is 29.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2010-II-Q13] | Part C (Q11-Q15)

Answer: 263

1. The other two cards must both lie below a or both above $a+9$. Thus $p(a)=[\binom{a-1}{2}+\binom{43-a}{2}]/\binom{50}{2}$, with $1\le a\le 43$.

2. The numerator equals $(a-22)^2+420$. The condition is $(a-22)^2\ge 385/2$, or $|a-22|\ge 14$ for integers.

3. The least admissible probability occurs at $a=8,36$, giving $616/1225=88/175$. Its numerator-denominator sum is 263.

Source / 来源

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2011-I-Q12] | Part C (Q11-Q15)

Answer: 594

1. The possible ordered male-block sizes under the conditioning are $(2,2,2),(2,4),(4,2),(3,3),(6)$. A pattern with j blocks occupies j of the $n+1$ female gaps.

2. The denominator count is $\binom{n+1}{3}+3\binom{n+1}{2}+n+1$. Favorable blocks are $(2,4),(4,2),(6)$, giving $2\binom{n+1}{2}+n+1$. Their ratio is $6(n+1)/(n^2+8n+6)$.

3. The one-percent condition is $n^2-592n-594\geq 0$. Its positive root lies strictly between 593 (value -1) and 594 (value 594); the other root is negative.

4. The least nonnegative integer number of women is therefore 594.

Source / 来源